

Abdelrahman Mahalawy

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Summary

Software engineer passionate about building well-structured, reliable applications and collaborating on products that solve real problems. Experienced across the full stack with a strong emphasis on backend development, system design, and practical delivery. Thrives in team environments, adapts quickly to new challenges, and is always looking to grow by contributing to ambitious engineering efforts alongside talented people.

Education

Information Technology Institute (ITI)

10/2025 – 08/2026

9-Month Diploma – Open-Source Applications Development

- Joined during military service deferment after being accepted among the top 4% of applicants, training in Node.js, NestJS, Next.js, Angular, MongoDB, MySQL, Design Patterns, Web Services, Docker, and Linux.

Ain Shams University, Faculty of Engineering (CESS)

06/2025

B.Sc. in Computer Engineering — Grade: Very Good

- Major: Distributed Computing | Minor: Software Product Lines
- Graduation Project: **EduAi** – Microservices-Based E-Learning Platform (Grade: Excellent)

Experience

Backend & Automation Developer, Freelance

2026 – Present

- Designed and delivered backend systems and data aggregation pipelines for multiple clients, consolidating data from external sources into structured databases and reporting layers.
- Engineered automated scheduling workflows using GitHub Actions to trigger recurring data collection jobs, storing results reliably in Google Sheets and Supabase.
- Built and deployed client-facing web applications integrating live aggregated data backends, handling data modeling, API design, and deployment end-to-end.

Internships

Software Developer Intern, I'SUPPLY — Cairo, Egypt

07/2023 – 10/2023

Cloud Computing Intern, Telecom Egypt – WE Digital Hub (D-HUB) — Cairo, Egypt

07/2023

Projects

EduAi, Microservices-Based E-Learning Platform

- Architected an e-learning platform using a microservices architecture, delivering core backend services for payments, user management, webhooks, and frontend interfaces.

wassalnii, Cross-Platform Mobile App

- Developed a real-time, offline-capable mobile app with Flutter, Firebase, and SQLite, featuring a responsive UI and seamless data synchronization.

Bash DBMS, Database Management System in Pure Bash

- Engineered a CLI-based DBMS entirely in Bash supporting full CRUD operations, typed columns, primary keys, and a lightweight SQL-like query interface over flat-file storage.

Mini-Fortran Compiler

- Engineered a FORTRAN grammar and implemented a mini-compiler in Python supporting lexical analysis, parse trees, error detection, and recovery.

Modified Minix3 OS

- Enhanced and recompiled the Minix3 kernel using C, deploying and testing custom system calls via WinSCP and Teraterm.

Skills

Languages: JavaScript, TypeScript, Python, Java, C/C++, Ruby, PHP, SQL (PostgreSQL, MySQL), Bash

Backend & Frameworks: Node.js, NestJS, Express.js, Laravel, Django, Ruby on Rails, REST APIs, Microservices

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Supabase

Frontend: React, Next.js, Angular, Vue, Bootstrap, TailwindCSS

Tools: AWS, GCP, Docker, Git, Postman, Jira, VS Code, Linux

Personal Skills: Problem-Solving, Communication, Teamwork, Agile Collaboration, Adaptability, Time Management

Spoken Languages: Arabic (Native), English (Professional Working Proficiency)

Additional Information

Nationality: Egyptian

Military Status: Service Completed